USN | 4 | C | B | | | C | S | | | | Course Code: **18CS34**

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING, CANARA ENGINEERING COLLEGE

Programme: **B.E. (CS&E)** Semester: **III** **CIE: TEST - II** Month: **October 2019**

Course Title: **COMPUTER ORGANISATION**

Duration: **1 HOUR 30 MINUTE** Max. Marks: **50**

Note: *Answer any* ***ONE*** *full question from each part*

Q. No.	Question	Marks	CO	RBTL
	PART - A			
1.a	Explain USB tree structure and an output operation of USB.	7	2	2
1.b	Explain synchronous bus communication with detailed timing diagram for input transfer.	8	2	2
1.c	What is the burst mode that PCI bus supports? Explain function of all the data transfer signals on PCI bus.	5	2	2
	OR			
2.a	Define direct memory access? Explain bus arbitration.	7	2	2
2.b	Explain asynchronous bus communication with detailed timing diagram for input operation and output operation.	8	2	2
2.c	Explain function of different categories SCSI bus signals.	5	2	2
	PART - B			
3.a	Explain internal organization of 2Mx8 dynamic memory chip.	8	3	2
3.b	Differentiate static and dynamic RAM technology.	5	3	2
3.c	Explain memory hierarchy with respect to speed, size and cost.	5	3	2
3.d	Explain cache-mapping functions with example.	12	3	2
	OR			
4.a	Explain internal structure of synchronous DRAM. Also, define latency, bandwidth and double data rate RAMs.	8	3	2

4.b	Differentiate synchronous and asynchronous dynamic RAMs.	5	3	2
4.c	Explain different types of ROMs.	5	3	2
4.d	Explain following concepts with respect to improving performance of memory. a) Write buffer b) Prefetching c) Lookup-free cache	12	3	2